

UNIT-1

I. Number Systems — Oral Questions

1. Convert the binary number **1101** into decimal – explain orally.
2. Convert **127 (octal)** to binary.
3. What is the decimal value of hexadecimal **2F**?
4. What is the binary equivalent of decimal **25**?
5. Convert **10011011 (binary)** to hexadecimal.
6. Convert decimal **345** to octal.
7. Convert hexadecimal **A3** to binary.
8. Convert **111101 (binary)** to octal.
9. Convert decimal **255** to hexadecimal.
10. Convert hexadecimal **1C** to decimal.
11. What is the octal equivalent of binary **101101**?

II. Binary Codes — Oral Questions

12. What is the **8421 BCD code** for decimal **9**?
13. How do you represent decimal **27** in BCD?
14. What do you mean by **Binary Coded Decimal (BCD)**?
15. Why do some digital systems use **BCD instead of binary**?
16. What is the difference between **binary** and **BCD** representation?
17. Convert decimal **45** into BCD.
18. How many bits are needed to represent **one decimal digit** in BCD?
19. Which **digit combinations** are invalid in BCD?
20. Why is **BCD inefficient** for arithmetic operations?
21. What are the **advantages of binary codes** in digital electronics?

III. Boolean Algebra & Theorems — Oral Questions

- 22. Write **De Morgan's theorems**.
- 23. What is the **dual** of the Boolean expression $AB + AC$?

IV. SOP, POS, Minterms, Maxterms — Oral Questions

- 24. What is the difference between **SOP** and **POS** forms?
- 25. Express $F(A,B,C) = \Sigma(1,3,5,7)$ in SOP (or orally explain what it means).
- 26. Write the **maxterm form** for $F = \Pi(0,2,4,6)$.
- 27. Define **minterm** and **maxterm** with one example each.

V. K-Maps & Minimization — Oral Questions

- 28. Draw and explain a **3-variable K-map** with proper cell labeling.
- 29. Draw and explain a **4-variable K-map** with cell addresses.
- 30. What is the **advantage of K-map** over Boolean algebra?
- 31. How do you **group terms** in a 4-variable K-map? Explain the rules.
- 32. What is a **Don't-care condition**? Give an example.
- 33. Draw K-map for following combinations:
 $F = \Sigma(0,5,9,12,13,14,15) + d(1,2,3,4)$
- 34. What is meant by **logic minimization**?
- 35. List the **steps** for minimizing a Boolean function using K-map.
- 36. How many **groups** are possible in a 3-variable K-map?
- 37. What is the **largest group size** in a 4-variable K-map?
- 38. Give **one real-life application** of logic minimization.

VI. Other Binary Codes — Gray, Excess-3, Conversions

- 39. What is the **Gray code** equivalent of binary **1010**?

- 40. Convert decimal **7** to **Excess-3** code.
- 41. What is the **purpose** of using Gray code?
- 42. How is **Gray code different** from binary code?
- 43. Convert **Gray code 1101** to binary.
- 44. What is the **main application** of Gray code?
- 45. What is the rule for converting **binary to Excess-3**?
- 46. What is **reflected binary code**?
- 47. How do you convert **BCD to Excess-3**?

VII. Binary Arithmetic, Complements, Subtraction

- 48. Add **1011 + 1101** in binary.
- 49. What is the sum of **1111 + 0001**?
- 50. Perform binary addition: **100 + 110**.
- 51. Explain how **carry** works in binary addition with an example.
- 52. State the **rules of binary addition**.
- 53. Subtract **1010 from 1101**.
- 54. Perform **1000 – 0011** in binary.
- 55. Subtract **0110 from 1001** using direct subtraction.
- 56. State the **rules of binary subtraction**.
- 57. What is the **1's complement** of binary **1101**?
- 58. Find the 1's complement of **00001111**.
- 59. Represent decimal **–9** in 1's complement (4-bit).
- 60. Perform **1011 – 0110** using 1's complement.
- 61. What is **end-around carry** in 1's complement subtraction?

UNIT-2

I. Multiplexers (MUX)

2:1 Multiplexer – Oral Questions

1. What is a **2:1 multiplexer**?
2. Explain the **block diagram** of a 2:1 MUX.
3. What is the **truth table** of a 2:1 multiplexer?
4. Give **two applications** of a 2:1 multiplexer.

4:1 Multiplexer – Oral Questions

5. What is a **4:1 multiplexer**?
6. How many **select lines** are required for a 4:1 MUX?
7. Explain the **internal logic diagram** of a 4:1 MUX.
8. State the **truth table** for a 4:1 MUX.
9. Give **two applications** of a 4:1 multiplexer.

8:1 Multiplexer – Oral Questions

10. What is an **8:1 multiplexer**?
11. How many **select lines** are used in an 8:1 MUX?
12. Explain the **logic diagram** of an 8:1 MUX.
13. State any **uses** of an 8:1 MUX in digital systems.

16:1 Multiplexer – Oral Questions

14. What is a **16:1 multiplexer**?
15. How many **select lines** are needed for a 16:1 MUX?
16. Explain the **working** of a 16:1 multiplexer.
17. Where is a **16:1 multiplexer** used?

Cascading Multiplexers

18. How do you **design a 16:1 multiplexer using four 4:1 multiplexers**?
19. What is meant by **cascading of multiplexers**?

Types of Multiplexers & Advantages — Oral Questions

20. What are the **different types** of multiplexers?
21. Explain the **advantages** of using multiplexers in digital circuits.

II. Adders and Subtractors

Half Adder – Oral Questions

22. What is a **half adder**?
23. State the **truth table** of the half adder.
24. What is the **limitation** of a half adder?

Full Adder – Oral Questions

25. What is a **full adder**?
26. How is a full adder **different** from a half adder?
27. What is the **truth table** of a full adder?
28. How can a full adder be designed using **two half adders**?

Half Subtractor – Oral Questions

29. What is a **half subtractor**?
30. Draw and explain its **logic diagram**.
31. State the **truth table** of a half subtractor.

Full Subtractor – Oral Questions

- 32. What is a **full subtractor**?
- 33. Explain its **logic diagram** using basic gates.
- 34. What is the **truth table** of a full subtractor?
- 35. Write the Boolean expressions for **Difference** and **Borrow**.
- 36. How is a full subtractor implemented using **two half subtractors**?

III. Decoders

2-to-4 Decoder – Oral Questions

- 37. Mention **applications** of decoders.
- 38. How is a decoder different from a **demultiplexer**?

IV. Parity Circuits

Parity Generator / Checker – Oral Questions

- 39. What is a **parity bit**?
- 40. What is the difference between **even parity** and **odd parity**?
- 41. How does a **parity checker** detect errors?
- 42. Where are parity circuits used in real life?

Difference-Based Oral Questions

- 43. What is the difference between a multiplexer and a demultiplexer
- 44. What is the difference between a decoder and a demultiplexer
- 45. What is the difference between MUX and decoder?
- 46. Where is DEMUX preferred over MUX in digital systems?

UNIT-3

I. Basics of Flip-Flops

1. What is a **flip-flop**?
2. Why is a flip-flop called a **1-bit memory cell**?
3. What is the difference between **combinational** and **sequential** circuits?
4. What is the function of a **clock signal** in flip-flops?

II. S-R Flip-Flop

5. Explain the **S-R latch** using its logic diagram.
6. What is the **truth table** of an S-R flip-flop?
7. Why is the **S = 1, R = 1** condition considered invalid?
8. What is the difference between **S-R latch** and **S-R flip-flop**?

III. J-K Flip-Flop

9. Draw and explain the **logic diagram** of a J-K flip-flop.
10. What are the **four modes** of operation of J-K flip-flop?
11. Explain **toggle**, **reset**, **set**, and **no-change** states of J-K FF.
12. What is the **race-around condition** in a J-K flip-flop?
13. Why does race-around occur only in **level-triggered** J-K FFs?
14. How does the **Master-Slave J-K flip-flop** solve the race-around problem?

IV. D Flip-Flop

15. Explain the working of a **D flip-flop**.
16. What is the **truth table** of a D flip-flop?
17. What is the **excitation table** of a D flip-flop?
18. Why is the D FF also called a **delay** flip-flop?

V. T Flip-Flop

19. Explain the working of a **T flip-flop**.
20. State the **truth table** of a T flip-flop.
21. Where is a T flip-flop mainly used?

VI. Characteristics of Flip-Flops

22. What are the **characteristics** of flip-flops?
23. What is **propagation delay**?

VIII. Registers

24. What is a **register**? Explain types of registers.
25. What is a **bidirectional shift register**?
26. What is a **ring counter** and **Johnson counter**?
27. What are the **applications** of shift registers?

IX. Counters – Asynchronous (Ripple)

28. What is an **asynchronous counter**?
29. Why is it called a **ripple counter**?
30. Draw the **state diagram** for the 4-bit asynchronous UP counter.

X. Counters – Synchronous

31. What is a **synchronous counter**?
32. Compare **synchronous** and **asynchronous** counters.
33. Why are synchronous counters faster?

XII. Excitation Tables & Design Importance

34. What is an **excitation table**?
35. Why are excitation tables important in **counter design**?
36. How are excitation tables used in **state machine design**?

XIII. Standard IC Counters

IC 7490

37. What is the use of **IC 7490**.
38. Explain applications of **IC 7490**.
39. What is the use of **IC 7493**.
40. Explain applications of **IC 7493**.

UNIT-4

I. Architecture & Organization

1. What is the function of the **Bus Interface Unit (BIU)** in 8086?
2. What is the function of the **Execution Unit (EU)** in 8086?
3. Evaluate: How do **EU and BIU work in parallel** to improve performance?
4. How is the physical address generated from **segment** and **offset**?
5. Define **segmentation** in 8086.

II. Registers

6. List the **different types of registers** in 8086.
7. Explain the role of **general-purpose registers**.
8. Explain the role of **segment registers**.
9. How do segment registers differ from **general-purpose registers**?

10. Show how **AX, BX, CX, and DX** can be split into **8-bit and 16-bit** parts.
11. Give examples of accessing **AL, AH, BL, BH, CL, CH, DL, DH**.
12. Explain the **flag register** of 8086 in detail.
13. What is the purpose of **status flags** and **control flags**?

III. Addressing Modes

A. Immediate, Direct, Register

14. Explain **Immediate addressing** with examples.
15. Explain **Direct addressing** with examples.
16. Explain **Register addressing** with examples.

B. Register Indirect, Indexed, Relative

17. Explain **Register Indirect addressing** with examples.
18. Explain **Indexed addressing** with examples.
19. Explain **Relative addressing** with examples.

C. Base, Based Indexed, String Addressing

20. Explain **Base addressing** with examples.
21. Explain **Based Indexed addressing** with examples.
22. Explain **String addressing mode** with examples.

D. I/O Addressing Modes

23. Explain **Direct I/O port addressing** with examples.
24. Explain **Indirect I/O port addressing** with examples.
25. Explain **Implied addressing** with suitable examples.

IV. Instruction Set

26. List the **different types of instructions** supported by 8086.
27. Explain the **general format** of an 8086 instruction.

A. Data Transfer Instructions

- 28. Explain **MOV** with examples.
- 29. Explain **XCHG**, **LEA**, and **PUSH/POP** with examples.

B. Arithmetic Instructions

- 30. Explain **ADD**, **SUB**, **MUL**, **DIV** with examples.
- 31. Explain **INC** and **DEC** instructions.

C. Branch Instructions

- 32. Explain **JMP** and **CALL** with examples.
- 33. Explain **conditional jump** instructions with examples.

D. Logical Instructions

- 34. Explain **AND**, **OR**, **XOR** with examples.
- 35. Explain **NOT** and **TEST** instructions.

E. Bit Manipulation Instructions

- 36. Explain **SHL**, **SHR** with examples.
- 37. Explain **ROL**, **ROR** with examples.

V. Pin Diagram & Bus System

A. Multiplexed Bus

- 38. Explain the function of the **multiplexed address/data bus (AD0–AD15)**.

B. Minimum Mode vs Maximum Mode

- 39. Differentiate between **Minimum Mode** and **Maximum Mode** operation of 8086.
- 40. Explain **control signals** generated in minimum mode.
- 41. Explain **control signals** generated in maximum mode.

VI. Pin Descriptions

Group 1: Basic Buses

42. Explain the pins **AD0–AD15**, **BHE'/S7**, **READY**, **RD'**.

Group 2: Interrupt & Control Pins

43. Explain **RESET**, **INTR**, **NMI**, **TEST** pins.

Group 3: Power & Clock Pins

44. Explain **CLK**, **VCC**, **INTA'**, **ALE**.

Group 4: Bus Control & DMA Pins

45. Explain the pins **LOCK'**, **RQ'/GT'**, **HOLD**, **HLDA**.

Group 5: Read/Write Mode Signals

46. Explain **WR'**, **MN/MX'**, **DEN'**, **DT/R'** signals.

UNIT-5

A. Introduction to Pentium

1. What is the Pentium processor? Why is it considered an important milestone in Intel's processor evolution?

B. CPU Pin Description

2. What roles do the address pins, data pins, and control pins play in Pentium?
3. During a hardware interrupt, how do interrupt-related pins function in the Pentium processor?
4. Can you differentiate between system pins and control pins with suitable examples?

C. RISC Concepts in Pentium

5. What is RISC? Why is the Pentium *not* considered a pure RISC processor?
6. Compare and contrast RISC and CISC characteristics present in the Pentium processor.

D. Bus Operations

7. What do you mean by bus operations in a Pentium processor?
8. What are the different types of bus cycles supported by Pentium?

E. Pentium's Superscalar Architecture

9. What is meant by superscalar architecture?
10. Compare Pentium's U and V pipelines with a traditional scalar pipeline.

F. Pipelining

11. What is instruction pipelining in Pentium?
12. What are the major stages of the Pentium pipeline? Explain their roles.

G. Branch Prediction

13. What is branch prediction? Why is it essential in Pentium?

H. Instruction & Data Caches

14. What is the purpose of instruction and data caches in Pentium?

UNIT-6

A. Interrupt Structure

1. What is an interrupt? What are the different types of interrupts in microprocessors?
2. Explain the general interrupt structure of the 8086 microprocessor.

B. Interrupt Service Routine (ISR) & Vector Table

3. What is an Interrupt Service Routine (ISR)?
4. Explain the sequence of steps that occur when the 8086 executes an ISR.

C. Hardware & Software Interrupts

5. Explain the difference between hardware interrupts and software interrupts, with examples.

D. INTR and NMI Interrupt Response

6. What are INTR and NMI interrupts in the 8086?
7. Compare maskable and non-maskable interrupts and their importance in safety-critical applications.

E. Execution of ISR & Priority of Interrupts

8. Mention the steps followed by the 8086 during ISR execution.

F. Programmable Interrupt Controller (8259) – Control Words

9. What is the function of ICW1, ICW2, ICW3, and ICW4 in the 8259?
10. Describe the purpose of OCW1, OCW2, and OCW3 in the 8259.